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School of Engineering and Technology
Department of Computer Science and Engineering
Experiment 4

Course Outcome (CO1): Understand the machine learning algorithms
BTL: L3
Objective: Implement different preprocessing operations on the given dataset.
Dataset: Titanic Dataset

Part A – Data Understanding

1. Load the Titanic dataset and inspect its structure by displaying the first few records, dimensions, column names, data types, and summary statistics.
2. Identify the missing values in the Titanic dataset and display the number and percentage of missing values in each column.
3. Handle missing values in the age column using mean and median imputation and compare the results.
4. Handle missing values in the embarked column using mode imputation and verify the result.
5. Remove rows and columns containing missing values from the Titanic dataset and compare the dataset dimensions before and after deletion.
6. Identify and remove duplicate records from the Titanic dataset and compare the dataset before and after removal.
7. Identify and separate numerical and categorical attributes from the Titanic dataset.

Part B – Data Transformation

8. Apply Min-Max normalization to the age and fare attributes of the Titanic dataset.
9. Standardize the age and fare attributes using Z-score standardization and display the transformed data.
10. Apply Robust Scaling to the age and fare attributes and observe the effect of scaling.
11. Detect outliers in the fare attribute using the Interquartile Range (IQR) method.
12. Treat the outliers in the fare attribute using the IQR-based capping method and compare the results.
13. Convert the sex attribute into binary numerical values using an appropriate encoding technique.
14. Apply One-Hot Encoding to the class attribute of the Titanic dataset and display the transformed data.

15. Apply One-Hot Encoding with drop_first=True to the class and embarked attributes.
16. Apply Label Encoding to the class attribute and display the encoded values and corresponding class labels.